

2 Metric spaces

Definition 2.3. Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f: X \rightarrow Y$ is called an isometric embedding if

$$d_Y(f(x), f(x')) = d_X(x, x') \quad \text{for all } x, x' \in X.$$

The function f is called an isometry if it is an isometric embedding that is also surjective and we say that (X, d_X) and (Y, d_Y) are *isometric* if there exists an isometry between them.

Example 2.4 (Isometric embedding). Let $f: \mathbb{R} \rightarrow \mathbb{R}^2$ be given by $f(t) = (t, 0)$. For $s, t \in \mathbb{R}$,

$$\|f(s) - f(t)\|_2 = \sqrt{(s-t)^2 + 0^2} = |s-t|.$$

Thus f preserves distances (an isometric embedding). It is not surjective, so it is not an isometry. Its image is the x -axis in $\mathbb{R}^2$.

Is the map $g(x) := (x, x)$ an isometric embedding?

Definition 2.5. Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f: X \rightarrow Y$ is called *Lipschitz continuous* if there exists a constant $L \geq 0$ such that

$$d_Y(f(x), f(x')) \leq L \cdot d_X(x, x') \quad \text{for all } x, x' \in X.$$

Definition 2.6. Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f: X \rightarrow Y$ is called *continuous* at a point $x_0 \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_Y(f(x), f(x_0)) < \varepsilon \quad \text{whenever } d_X(x, x_0) < \delta.$$

The function f is called *continuous* if it is continuous at every point $x_0 \in X$.

Proposition 2.7. Any Lipschitz continuous function between metric spaces is continuous. Any isometric embedding is also Lipschitz continuous.

Proof. Let $f: X \rightarrow Y$ be Lipschitz continuous with constant L . Given $\varepsilon > 0$, let $\delta = \varepsilon/L$. Then, for any $x \in X$ with $d_X(x, x_0) < \delta$, we have

$$d_Y(f(x), f(x_0)) \leq L \cdot d_X(x, x_0) < L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

Thus, f is continuous at x_0 . Since x_0 is arbitrary, f is continuous.

If f is an isometric embedding, then for any $x, x' \in X$, we have

$$d_Y(f(x), f(x')) = d_X(x, x') \leq 1 \cdot d_X(x, x').$$

Thus, f is Lipschitz continuous with constant $L = 1$. □

2.1 Characterizations of continuity

2.1.1 A sequential characterization of continuity

For real functions, we can define continuity using sequences. This can be generalized to metric spaces as well.

Definition 2.8. A sequence $(x_n)_{n \in \mathbb{N}}$ in a metric space (X, d) is said to converge to a point $x \in X$ if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_n, x) < \epsilon$ for all $n \geq N$.

One can characterize continuity using sequences:

Proposition 2.9. A function $f : X \rightarrow Y$ is continuous at a point $x_0 \in X$ if and only if for every sequence $(x_n)_{n \in \mathbb{N}}$ in X converging to x_0 , the sequence $(f(x_n))_{n \in \mathbb{N}}$ converges to $f(x_0)$.

Proof. Suppose first that f is continuous at x_0 , and let $(x_n)_{n \in \mathbb{N}}$ be a sequence converging to x_0 . For any $\varepsilon > 0$, by continuity there exists $\delta > 0$ such that

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

Since (x_n) converges to x_0 , there exists an $N \in \mathbb{N}$ with $d_X(x_n, x_0) < \delta$ for all $n \geq N$. Hence, for all $n \geq N$, we have

$$d_Y(f(x_n), f(x_0)) < \varepsilon,$$

so $(f(x_n))$ converges to $f(x_0)$.

Conversely, suppose that for every sequence (x_n) converging to x_0 , the sequence $(f(x_n))$ converges to $f(x_0)$. Assume for contradiction that f is not continuous at x_0 . Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exists some $x \in X$ with

$$d_X(x, x_0) < \delta \quad \text{but} \quad d_Y(f(x), f(x_0)) \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choosing $\delta = \frac{1}{n}$, there exists $x_n \in X$ with

$$d_X(x_n, x_0) < \frac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(x_0)) \geq \varepsilon_0.$$

Then (x_n) converges to x_0 , yet $(f(x_n))$ cannot converge to $f(x_0)$ since the distance remains at least ε_0 , which is a contradiction. $\square$

2.1.2 A open set characterization of continuity

Definition 2.10. Let (X, d) be a metric space. For any $x \in X$ and $r > 0$, the open ball of radius r centered at x is defined as

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

Example 2.11 (Shapes of balls for different metrics on $\mathbb{R}^2$). For $r > 0$ the open unit balls (centered at the origin) for the standard metrics on $\mathbb{R}^2$ have the following shapes:

$$\begin{aligned} B_r^{(2)}(0) &= \{x \in \mathbb{R}^2 : \|x\|_2 < r\} && \text{(Euclidean) — disk,} \\ B_r^{(1)}(0) &= \{x \in \mathbb{R}^2 : |x_1| + |x_2| < r\} && \text{(Taxicab / } \ell^1) \text{ — diamond,} \\ B_r^{(\infty)}(0) &= \{x \in \mathbb{R}^2 : \max(|x_1|, |x_2|) < r\} && \text{(Maximum / } \ell^\infty) \text{ — square.} \end{aligned}$$

Below we draw the case $r = 1$.

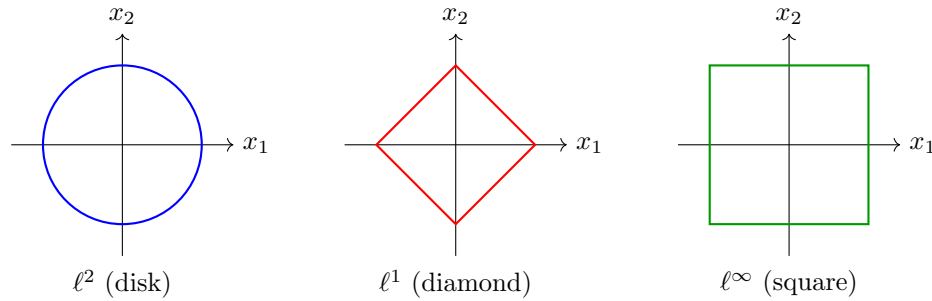


Figure 1: Unit balls $B_1(0)$ in $\mathbb{R}^2$ for ℓ^2 , ℓ^1 , ℓ^∞ metrics.

Then, we can rewrite the definition of continuity using open balls.

Proposition 2.12. *A function $f : X \rightarrow Y$ between metric spaces (X, d_X) and (Y, d_Y) is continuous at a point $x_0 \in X$ if and only if for every $r > 0$, there exists $\delta > 0$ such that*

$$f(B_\delta(x_0)) \subseteq B_r(f(x_0)).$$

The definition of open balls of course depends on the metric. Suppose you want to generalize everything so far to the setting when you don't have metric. Then, let's try to define some language where the notion of metric will be hidden.

Definition 2.13. A subset $U \subseteq X$ is called *open* if for every point $x \in U$ there exists a radius $r > 0$ such that the open ball $B_r(x)$ is contained in U .

Proposition 2.14. *Any open ball $B_r(x)$ is open.*

Proof. Let's prove the last item. Let $x \in B_r(x_0)$ for some $r > 0$. Pick $r' := r - d_X(x, x_0)$. Then, $B_{r'}(x) \subseteq B_r(x_0)$. $\square$

Corollary 2.15. *Let (X, d) be a metric space and $U \subseteq X$ be a subset. Then, U is open iff U is a union of open balls.*

Proof. Assume that U is open. Then, for every $x \in U$ there exists $r_x > 0$ so that $B_{r_x}(x) \subseteq U$. Thus, we can write

$$U = \bigcup_{x \in U} B_{r_x}(x).$$

Conversely, if $U = \bigcup_{i \in I} B_{r_i}(x_i)$ is a union of open balls, then for every $x \in U$, we have that $x \in B_{r_i}(x_i)$ for some $i \in I$. This means that there exists a radius $r > 0$ such that $B_r(x) \subseteq B_{r_i}(x_i) \subseteq U$. Thus, U is open. $\square$

Now we can revisit the definition of continuous functions.

Theorem 2.16. *Let $f : X \rightarrow Y$ be a function between metric spaces (X, d_X) and (Y, d_Y) . Then f is continuous at a point $x_0 \in X$ if and only if for every open set $V \subseteq Y$ containing $f(x_0)$, there exists an open set $U \subseteq X$ containing x_0 such that $f(U) \subseteq V$.*

Proof. Suppose that f is continuous at x_0 . Let $V \subseteq Y$ be an open set containing $f(x_0)$. By openness, there exists $r > 0$ such that $B_r(f(x_0)) \subseteq V$. By continuity, there exists $\delta > 0$ such that $f(B_\delta(x_0)) \subseteq B_r(f(x_0)) \subseteq V$. Thus, we can take $U = B_\delta(x_0)$.

Conversely, suppose that for every open set $V \subseteq Y$ containing $f(x_0)$, there exists an open set $U \subseteq X$ containing x_0 such that $f(U) \subseteq V$. Given $\varepsilon > 0$, let $V = B_\varepsilon(f(x_0))$. By assumption, there exists an open set $U \subseteq X$ containing x_0 such that $f(U) \subseteq V$. Since U is open and contains x_0 , there exists $\delta > 0$ such that $B_\delta(x_0) \subseteq U$. Thus,

$$f(B_\delta(x_0)) \subseteq f(U) \subseteq V = B_\varepsilon(f(x_0)).$$

This shows that f is continuous at x_0 . $\square$

Finally, we will be able to establish the following characterization from the theorem above. We will not give a proof here as we will give a proof in a more general setting (topological spaces) later.

Theorem 2.17. *f is continuous if and only if for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X .*

The above theorem reads "preimage of open is open".

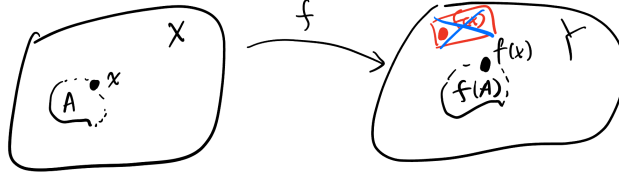
2.1.3 A closed set characterization of continuity

However, I never find this characterization natural to understand. We can think of the continuity as not ripping apart the space at any given point. More precisely, if a point is "in touch" with some region in the domain, it should remain in touch with the corresponding region in the codomain.

Definition 2.18. A subset $A \subseteq X$ is called *closed* if its complement $X \setminus A$ is open.

Definition 2.19. For any set $A \subseteq X$, the *closure* of A , denoted $\overline{A}$, is the smallest closed set containing A . Equivalently, it is the intersection of all closed sets containing A .

Theorem 2.20. *Let $f : X \rightarrow Y$ be a function between metric spaces (X, d_X) and (Y, d_Y) . Then f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$.*



The theorem above reads "image of closure is contained in closure of image", or more intuitively, "points that are in touch with a set remains in touch after the map".

Now we prove the theorem.

Lemma 2.21. *Let $A \subseteq X$ be a set. Then,*

$$\overline{A} = \{x \in X : \text{there exists a sequence } (x_n)_{n \in \mathbb{N}} \text{ in } A \text{ converging to } x\}.$$

Proof. Let $x \in \overline{A}$. Consider any $r > 0$. If $B_r(x) \cap A = \emptyset$, then the complement $X \setminus B_r(x) \supset A$. But this contradicts the fact that $x \in \overline{A}$. Now let $r_n := 1/n$. Then, pick $x_n \in B_{r_n}(x) \cap A$. Then, (x_n) is a sequence in A converging to x .

Conversely, let $x \in X$ be such that there exists a sequence $(x_n)_{n \in \mathbb{N}}$ in A converging to x . Consider a closed set $C \supset A$. If $x \notin C$, then there exists $r > 0$ such that $B_r(x) \cap C = \emptyset$ (why?). This implies that $B_r(x) \cap A = \emptyset$ which contradicts the fact that (x_n) converges to x . Thus, $x \in C$. Since $C \supset A$ is arbitrary, we have $x \in \overline{A}$. $\square$

Now, we prove Theorem 2.20.

Proof of Theorem 2.20. Suppose that f is continuous. Let $x \in \overline{A}$. By the lemma, there exists a sequence (x_n) in A converging to x . By continuity, $(f(x_n))$ converges to $f(x)$. Since each $f(x_n) \in f(A)$, by the lemma again, we have $f(x) \in \overline{f(A)}$. Thus, $f(\overline{A}) \subseteq \overline{f(A)}$.

Conversely, suppose that $f(\overline{A}) \subseteq \overline{f(A)}$ for every set $A \subseteq X$. Let $x_0 \in X$ and let (x_n) be a sequence converging to x_0 . Assume on the contrary that $(f(x_n))$ does not converge to $f(x_0)$. Then, there exists $\varepsilon > 0$ such that there is a subsequence $(f(x_{k_n}))$ such that $d_Y(f(x_{k_n}), f(x_0)) \geq \varepsilon$ for all $n \in \mathbb{N}$.

Now we let $A = \{x_{k_n} : n \in \mathbb{N}\}$. Then, $x_0 \in \overline{A}$. Thus, $f(x_0) \in \overline{f(A)} \subseteq \overline{f(A)}$. By the lemma, there exists a sequence (y_n) in $f(A)$ converging to $f(x_0)$. But this is not possible due to the ε constraint. Therefore, we conclude that $(f(x_n))$ must converge to $f(x_0)$. $\square$

We now see that many intuitive properties of continuous real functions can be generalized nicely to metric spaces. We will see that the open and closed language still works well in the more general setting of topological spaces, while the sequential language does not. In Willard's book, a whole chapter is devoted to the study of sequences in topological spaces.

2.2 Ultrametric

There is a special class of metric spaces called ultrametric spaces, which satisfy a stronger version of the triangle inequality. Ultrametric spaces have some interesting properties that are not shared by general metric spaces.

Definition 2.22 (Ultrametric space). A metric space (X, d) is an *ultrametric space* if d satisfies the *strong triangle inequality*

$$d(x, z) \leq \max\{d(x, y), d(y, z)\} \quad \text{for all } x, y, z \in X.$$

We would usually use u instead of d to denote an ultrametric.

Proposition 2.23. *Let (X, u) be an ultrametric space. Then,*

1. *If $r > 0$ and $x, y \in X$ with $u(x, y) < r$, then $B_r(x) = B_r(y)$.*
2. *Any two open balls are either disjoint or one contains the other.*
3. *Every triangle is isosceles, i.e., if $x, y, z \in X$, then at least two of the distances $u(x, y)$, $u(y, z)$, and $u(z, x)$ are equal, and the third is less than or equal to the common value.*

Example 2.24. Recall the definition of the Cantor set $C \subseteq [0, 1]$: it is constructed by repeatedly removing the open middle third from each interval. More precisely, set $C_0 = [0, 1]$, and for each $k \geq 1$, let C_k be the union of 2^k closed intervals obtained by removing the open middle third from each interval in C_{k-1} . The Cantor set is defined as

$$C = \bigcap_{k=0}^{\infty} C_k.$$

Equivalently, C consists of all numbers in $[0, 1]$ whose base-3 expansion contains only the digits 0 and 2. We endow the Cantor set $C \subseteq [0, 1]$ with the ultrametric as follows:

$$d_C(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 3^{-n} & \text{if } x \neq y \text{ and } n = \min\{k \geq 0 : x \text{ and } y \text{ lie in different segments in } C_k\}. \end{cases}$$

It is perhaps difficult for you to understand what happens in the example above. It turns out that one can always create a nice visualization of ultrametric spaces using trees. We will use this to help us understand ultrametric spaces better. Although the following definition is for finite sets, it can be generalized to infinite sets as well.

Definition 2.25. Dendrogram (finite case). Let X be finite. A dendrogram on X is an order preserving map $\theta : [0, \infty) \rightarrow \text{Part}(X)$ on X such that:

1. $\theta(0)$ is the discrete partition of X ;
2. there exists $T > 0$ such that $\theta(T)$ is the trivial partition;
3. for any $t \geq 0$, there exists $\epsilon > 0$ such that $\theta(t) = \theta(t + \epsilon)$.

Example 2.26. Consider $X = \{a, b, c\}$. Define $\theta : [0, \infty) \rightarrow \text{Part}(X)$ by

$$\theta(t) = \begin{cases} \{\{a\}, \{b\}, \{c\}\} & \text{if } 0 \leq t < 1 \\ \{\{a, b\}, \{c\}\} & \text{if } 1 \leq t < 2 \\ \{\{a, b, c\}\} & \text{if } t \geq 2 \end{cases}$$

Notice that the strict inequalities cannot be replaced with non-strict inequalities, as this would violate the last condition.

Theorem 2.27 (Finite dendrograms and ultrametrics are equivalent). *Let X be a finite set. Let $\mathcal{D}(X)$ be the set of dendrograms on X and $\mathcal{U}(X)$ be the set of ultrametrics on X . There are two maps between $\mathcal{D}(X)$ and $\mathcal{U}(X)$ as follows:*

1. (From dendrogram to ultrametric) Given a dendrogram θ on X , define

$$u(x, y) := \inf\{t \geq 0 : x, y \text{ belong to the same block in } \theta(t)\} \quad (x, y \in X).$$

Then u is an ultrametric on X .

2. (From ultrametric to dendrogram) Given an ultrametric u on X set, we define an equivalence relation $\sim_t$ for $t \geq 0$,

$$x \sim_t y \iff u(x, y) \leq t.$$

Then we define θ by $\theta(t) := \{[x]_t : x \in X\}$ where $[x]_t$ is the equivalence class of x under $\sim_t$. This gives a dendrogram on X .

The above two maps, denoted by $\mathcal{D} : \mathcal{D}(X) \rightarrow \mathcal{U}(X)$ and $\mathcal{U} : \mathcal{U}(X) \rightarrow \mathcal{D}(X)$ respectively, are inverses of each other.

Example 2.28. Consider the dendrogram θ on $X = \{a, b, c\}$ defined in the previous example. The corresponding ultrametric u is given by

$$u(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } \{x, y\} = \{a, b\}, \\ 2 & \text{if } \{x, y\} = \{a, c\} \text{ or } \{b, c\}. \end{cases}$$

Conversely, starting from this ultrametric u , we can recover the original dendrogram θ .

3 Topological spaces and continuous functions

This section introduces the notion of a topological space, which abstracts the concept of “openness” captured by metric spaces.

Definition 3.1 (Topological Space). A *topology* on a set X is a family τ of subsets of X (called *open sets*) satisfying:

1. Both $\emptyset$ and X belong to τ ;
2. Arbitrary unions of sets in τ belong to τ : if $\{U_i\}_{i \in I} \subseteq \tau$, then $\bigcup_{i \in I} U_i \in \tau$;
3. Finite intersections of sets in τ belong to τ : if $\{V_j\}_{j=1}^n \subseteq \tau$, then $\bigcap_{j=1}^n V_j \in \tau$.

The pair (X, τ) is called a *topological space*. In many cases, we will omit the mention of the topology and simply refer to the space X .

Definition 3.2. Let (X, τ) be a topological space. A subset $A \subseteq X$ is called *closed* if its complement $X \setminus A$ is open, i.e., $X \setminus A \in \tau$.

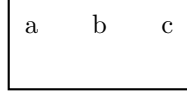
Proposition 3.3. 1. X and $\emptyset$ are closed.

2. Any finite union of closed sets is closed.

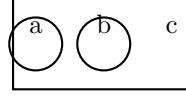
3. Any arbitrary intersection of closed sets is closed.

Remark 3.4. A topology, i.e., the collection of open sets, can be equivalently described by the collection of closed sets.

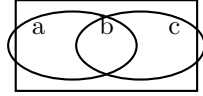
Example 3.5. Which of the following collections of subsets of $X = \{a, b, c\}$ are topologies?



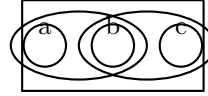
Yes: $\{\emptyset, X\}$ (trivial topology)



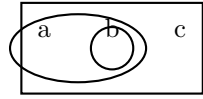
No



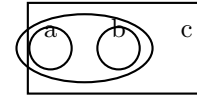
No



Yes



Yes



Yes

Example 3.6. Every metric space is a topological space with the open sets defined previously.

As a special example, for $\mathbb{R}$, every open ball is a bounded open interval (a, b) . Hence, every open set in $\mathbb{R}$ is a union of disjoint open intervals (including unbounded intervals).

Let (X, τ) be a topological space. If there exists a metric d on X such that open sets for d coincide with τ , we say (X, τ) is *metrizable*.

Example 3.7. Let X be any set. Let $\tau := P(X)$. Then, τ is a topology on X and it is called the *discrete topology*. This topology is induced by the discrete metric $d(x, y) = 1$ if $x \neq y$ and $d(x, x) = 0$.

Example 3.8. Let X be any set. Let $\tau := \{\emptyset, X\}$. This is the *trivial topology*. Is this metrizable?

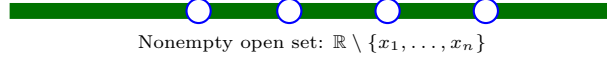
Lemma 3.9. For a metric space (X, d) , any point $x \in X$ forms a closed set, i.e., $\{x\}$ is closed.

Proof. We show that $X \setminus \{x\}$ is open. For any $y \in X \setminus \{x\}$, let $r = d(x, y)/2 > 0$. Then, the open ball $B(y, r) \subseteq X \setminus \{x\}$. Hence, $X \setminus \{x\}$ is open. $\square$

Now, for any set X with more than two elements, the trivial topology is not metrizable since singletons are not closed.

Example 3.10. Let X be any set. Let $\tau := \{O \subset X \mid O = \emptyset \text{ or } X \setminus O \text{ is finite}\}$. Then τ is a topology on X called the *finite complement topology*.

If $X = \mathbb{R}$, a nonempty open set is $\mathbb{R}$ with at most finitely many points removed. For example:



Proof. Now we check that the finite complement topology is indeed a topology.

- The empty set and the whole space X are in τ .
- The intersection of finitely many open sets is open: If $U_1, \dots, U_n \in \tau$ then each $X \setminus U_i$ is finite. Hence

$$X \setminus \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X \setminus U_i)$$

is a finite union of finite sets, hence finite. Therefore $\bigcap_{i=1}^n U_i \in \tau$.

- The union of any collection of open sets is open: If $\{U_i\}_{i \in I} \subseteq \tau$, then $X \setminus \bigcup_{i \in I} U_i = \bigcap_{i \in I} (X \setminus U_i) \subset X \setminus U_i$ is finite. Thus, $\bigcup_{i \in I} U_i \in \tau$.

□

The finite complement topology coincides with the trivial topology if X is finite and hence is metrizable. If X is infinite, the finite complement topology is not metrizable. We will need more machinery to show this.

Definition 3.11 (Comparing topologies). Let τ and τ' be topologies on the same set X . If $\tau \subseteq \tau'$ we say τ is *coarser* (or *weaker*) than τ' , and τ' is *finer* (or *stronger*) than τ .

Remark 3.12. The discrete topology $P(X)$ is the finest topology on X , while the trivial topology $\{\emptyset, X\}$ is the coarsest.

3.1 Basis

Recall that in a metric space, any open set can be written as a union of open balls. So the set of open balls gives rise to a simple characterization of the metric topology. We can use this idea to define a basis for a topology.

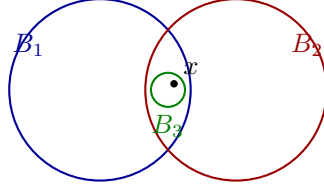
Definition 3.13 (Basis). A *basis* $\mathcal{B}$ for a topology on a set X is a nonempty family of subsets of X such that:

1. For every $x \in X$, there exists some $B \in \mathcal{B}$ such that $x \in B$.
2. If $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

The topology generated by $\mathcal{B}$ is: any $U \subset X$ is open if for every $x \in U$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$. Equivalently,

$$\tau_{\mathcal{B}} := \left\{ \bigcup_{i \in I} B_i : B_i \in \mathcal{B} \right\}.$$

We say “ $\mathcal{B}$ is a basis for $\tau_{\mathcal{B}}$ ”.



If $x \in B_1 \cap B_2$ there exists $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

Example 3.14. Let X be a metric space. Then, the collection of open balls forms a basis for the topology on X .

Example 3.15. The collection of one point sets $\{\{x\} : x \in X\}$ is a basis for the discrete topology on X .

Example 3.16. Let τ be a topology on X . Then, $\mathcal{B} := \tau$ is a basis for τ . This is a rather trivial case.

Proposition 3.17. *The topology generated by a basis is indeed a topology.*

Proof. • $\emptyset$ and X are in $\tau_{\mathcal{B}}$: $\emptyset = \bigcup_{i \in \emptyset} B_i$ and $X = \bigcup_{x \in X} B_x$ where $B_x \in \mathcal{B}$ contains x .

- Arbitrary unions of sets in $\tau_{\mathcal{B}}$ belong to $\tau_{\mathcal{B}}$: Consider $\{U_i\}_{i \in I} \subseteq \tau_{\mathcal{B}}$. Then, if $x \in \bigcup_{i \in I} U_i$, there exists $i_0 \in I$ such that $x \in U_{i_0}$. Since U_{i_0} is a union of basis elements, we can find $B \in \mathcal{B}$ with $x \in B \subseteq U_{i_0}$. Thus, $x \in B \subseteq \bigcup_{i \in I} U_i$. Hence, $\bigcup_{i \in I} U_i \in \tau_{\mathcal{B}}$.
- Now for $U_1, \dots, U_n \in \tau_{\mathcal{B}}$, let $x \in U_1 \cap \dots \cap U_n$. We claim that there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U_1 \cap \dots \cap U_n$. We prove by induction on n . The case $n = 1$ is trivial. Assume the claim holds for $n = k$. Now consider $n = k + 1$. By the induction hypothesis, there exists $B' \in \mathcal{B}$ such that $x \in B' \subseteq U_1 \cap \dots \cap U_k$. Since U_{k+1} is open, there exists $B'' \in \mathcal{B}$ such that $x \in B'' \subseteq U_{k+1}$. Now, by the definition of basis, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq B' \cap B''$. Hence, $x \in B \subseteq B' \cap B'' \subseteq (U_1 \cap \dots \cap U_k) \cap U_{k+1} = U_1 \cap \dots \cap U_{k+1}$. This proves the claim for $n = k + 1$ and hence for all n . $\square$

Proposition 3.18. *Let (X, τ) be a topological space and let $\mathcal{B} \subset \tau$. Assume that for any $x \in U$ where $U \in \tau$, there exists $B \in \mathcal{B}$ with $x \in B \subseteq U$. Then $\mathcal{B}$ is a basis for τ .*

Proof. By hypothesis, for any open $U \in \tau$ and any $x \in U$ there exists $B \in \mathcal{B}$ with $x \in B \subseteq U$.

(1) Take $U = X \in \tau$. Then for every $x \in X$ there is $B \in \mathcal{B}$ with $x \in B \subseteq X$, verifying the first basis axiom.

(2) If $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B} \subseteq \tau$, then $U := B_1 \cap B_2 \in \tau$. Applying the hypothesis to U and x yields $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$, which is the second basis axiom.

Thus $\mathcal{B}$ is a basis. Let $\tau_{\mathcal{B}}$ be the topology it generates. Since $\mathcal{B} \subseteq \tau$ and τ is closed under unions, $\tau_{\mathcal{B}} \subseteq \tau$. Conversely, for any $U \in \tau$,

$$U = \bigcup_{x \in U} B_x$$

where each $B_x \in \mathcal{B}$ satisfies $x \in B_x \subseteq U$ by the hypothesis. Hence $U \in \tau_{\mathcal{B}}$, so $\tau \subseteq \tau_{\mathcal{B}}$. Therefore $\tau = \tau_{\mathcal{B}}$, and $\mathcal{B}$ is a basis for τ . $\square$

Proposition 3.19 (Comparing topologies via bases). *Let $\mathcal{B}_1, \mathcal{B}_2$ be bases on X generating τ_1, τ_2 . Then $\tau_1 \subseteq \tau_2$ iff for every $x \in X$ and $B_1 \in \mathcal{B}_1$ with $x \in B_1$ there exists $B_2 \in \mathcal{B}_2$ with $x \in B_2 \subseteq B_1$.*

Proof. ($\Rightarrow$) If $B_1 \in \mathcal{B}_1$, then $B_1 \in \tau_1 \subseteq \tau_2$ and so B_1 is a union of $\mathcal{B}_2$ -elements. Pick one containing x inside B_1 .

($\Leftarrow$) Let $U \in \tau_1$; write $U = \bigcup_i B_i$ with $B_i \in \mathcal{B}_1$. For $x \in U$ choose i with $x \in B_i$ and then $B_2 \in \mathcal{B}_2$ with $x \in B_2 \subseteq B_i \subseteq U$. Hence U is open in τ_2 by the basis criterion. $\square$

Example 3.20. Let $X = \mathbb{R}^2$. Let $\mathcal{B} = \{\text{open balls in } \mathbb{R}^2\}$ and let $\mathcal{B}' = \{\text{axis-aligned open rectangles in } \mathbb{R}^2\}$. Then they generate the same topology.

